Launch vehicle conceptual design report

TSTO-LOX-RP1-10t-LEO — 2 stages

GLOM	PAYLOAD	ΔV REQUIRED	LENGTH
410,003 kg	**10,000 kg**	**9,408 m/s**	**54.87 m**
GROSS LIFT-OFF MASS	TO THE INJECTION ORBIT	INCL. LOSSES AND MARGIN	NOZZLE EXIT TO FAIRING TIP

MAX DIAMETER	LIFT-OFF T/W	PAYLOAD FRACTION
4.00 m	**1.30**	**2.44 %**
CORE / FAIRING	THRUST 5,227 KN	OF GLOM

Converged in 23 iterations (relative residual 8.18e-10, tolerance 1e-09). Achieved Δv 9,408 m/s against a requirement of 9,408 m/s.

Item	Value
Design	TSTO-LOX-RP1-10t-LEO
Configuration	2 stages
Mission	LEO, payload 10,000 kg
Design file	`config/default_design.yaml` sha256:05b24045b70d37b9
Assumptions file	`config/engineering_assumptions.yaml` sha256:40c6118a51bc62a6
Tool version	launch_vehicle 0.5.0
Generated	2026-10-12 20:51
Reference	Edberg & Costa, *Design of Rockets and Space Launch Vehicles*, 2nd ed., AIAA 2022 (with the published 1st-edition errata applied; see docs/ERRATA_COMPLIANCE.md)

This report is a **conceptual sizing** result produced by mass-estimating relationships and closed-form geometry. It is not a design qualification and does not demonstrate that the vehicle can fly. In particular:

- **No trajectory simulation.** Ascent losses are either a published regression fit or user estimates; the achievability of the orbit is not demonstrated, and no steering or lofting profile exists.
- **No structural analysis.** Tank and structure masses come from statistical MERs (or thin-wall membrane theory when that option is selected). Buckling, axial and bending loads, slosh, acoustics and proof testing are not sized. Nothing here is a stress report.
- **No stability, control, thermal, GNC, separation or aerodynamic analysis.** Moments of inertia are reported, but no control authority or margin is assessed.
- **MER scatter.** Mass-estimating relationships are fits to historical vehicles; individual components can deviate by tens of per cent. Validation against twelve real vehicles gives a median GLOM error of about 17 % (see docs/VALIDATION_REPORT.md). Treat every mass here as having that kind of uncertainty.
- **Engines and motors are rubber.** Unless a real engine's mass and dimensions were entered, the engine is a scaled MER and its envelope is a layout estimate.

Every input used is listed in full, with its source and provenance, in the **Input register** section; every calculation is listed in the **Engineering method register** with the book page it comes from or an explicit mark that it is supplementary.

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1. Mission requirement and Δv budget

Requirement	Value
Injection orbit	LEO: 400 km circular, i = 28.50 deg
Launch site	latitude 28.50 deg, altitude 0 m
Launch azimuth	computed from the inclination
Earth rotation credited	yes
Loss model	book_penalty — Eqs. 3:35-3:38 regression (lumped)
Design margin	0.0 %
Air launch	no (ground launch)
Δv override	none
Δv allocation	fractions → stage 1: 4,139.4 m/s, stage 2: 5,268.3 m/s

Δv budget, line by line

Budget line	m/s	Category	In total	Source
Ideal injection speed (inertial, at perigee)	7,671.7	ideal	yes	Eq. 3:1 / 3:4 (book pp.156-157)
Earth-rotation contribution (vector, launch site)	-408.7	rotation	yes	Book Example 3.9 (p.185) method; negative = benefit
Ascent losses: gravity + drag + steering + propulsion (lumped, Book Eq. 3:37-3:38)	2,144.6	loss	yes	Book regression pp.181-182: K3=1266.5, K4=1.9405, T_mix=452.5 s (T_ascent=548.4 s, T_3s=387.2 s)
indicative gravity share of lumped losses (Table 3.2 proportions) *(informational)*	1,784.6	info	no	Informational split only - NOT added to the total (already inside the lumped penalty)
indicative drag share of lumped losses (Table 3.2 proportions) *(informational)*	135.6	info	no	Informational split only - NOT added to the total (already inside the lumped penalty)
indicative steering share of lumped losses (Table 3.2 proportions) *(informational)*	134.4	info	no	Informational split only - NOT added to the total (already inside the lumped penalty)
indicative propulsion share of lumped losses (Table 3.2 proportions) *(informational)*	90.0	info	no	Informational split only - NOT added to the total (already inside the lumped penalty)
Other allowances (reserves, extra)	0.0	allowance	yes	User-defined
Design margin (0.0% of subtotal)	0.0	margin	yes	User-defined fraction applied once to the subtotal
Δv required	**9,407.6**	**total**		

2. Vehicle configuration

Stage	Name	Type	Propellant	Propulsion	D [m]	L [m]	Isp_eff [s]	Δv [m/s]	Propellant [kg]	Dry [kg]	s = ms/(ms+mp)
1	Stage 1	liquid	LOX/RP-1, O/F 2.50	6 x Generic LOX/RP-1 booster engine	4.00	33.88	290.0	4,139	322,638	14,156	0.0656
2	Stage 2	liquid	LOX/RP-1, O/F 2.50	1 x Generic LOX/RP-1 vacuum engine	3.00	12.92	340.0	5,268	59,037	2,661	0.0666

Upper assembly	Value
Payload	10,000 kg in a 2.70 m x 3.00 m envelope
Payload adapter (PAF)	805.0 kg (method: mer)
Fairing	705.3 kg, 3.00 m diameter, metal, jettisoned at the start of stage 2

3. Drawings

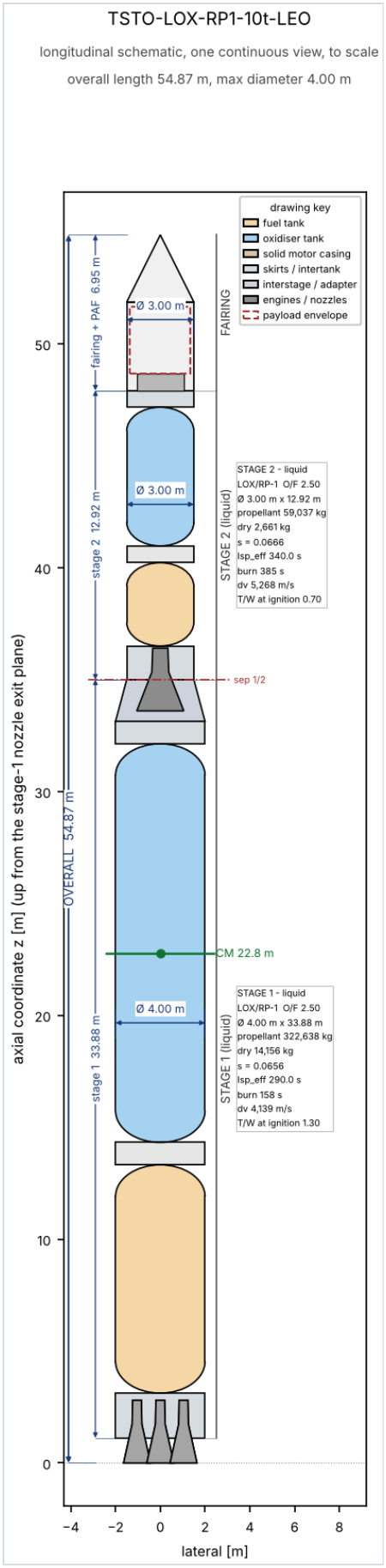


Figure 1. Vehicle general arrangement, to scale. The vehicle is drawn as one continuous view at the largest scale that fits the page height, with the dimensions on the left and the stage data beside the stage it describes. Interior outlines are the sized tank or motor-

case geometry, not a structural layout.

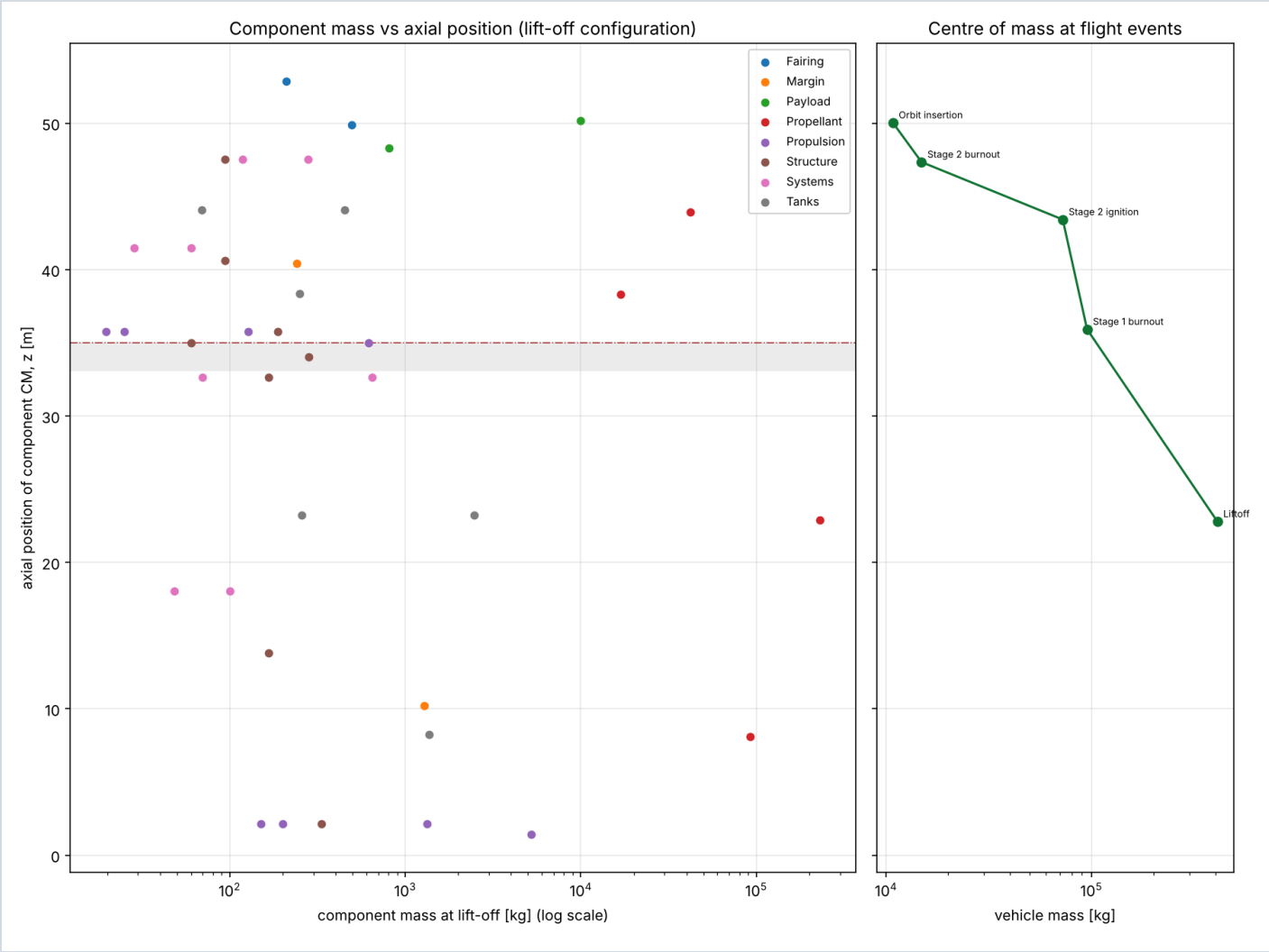


Figure 2. Mass distribution along the vehicle axis and the centre-of-mass travel through the flight events, from the mass ledger in section 6.

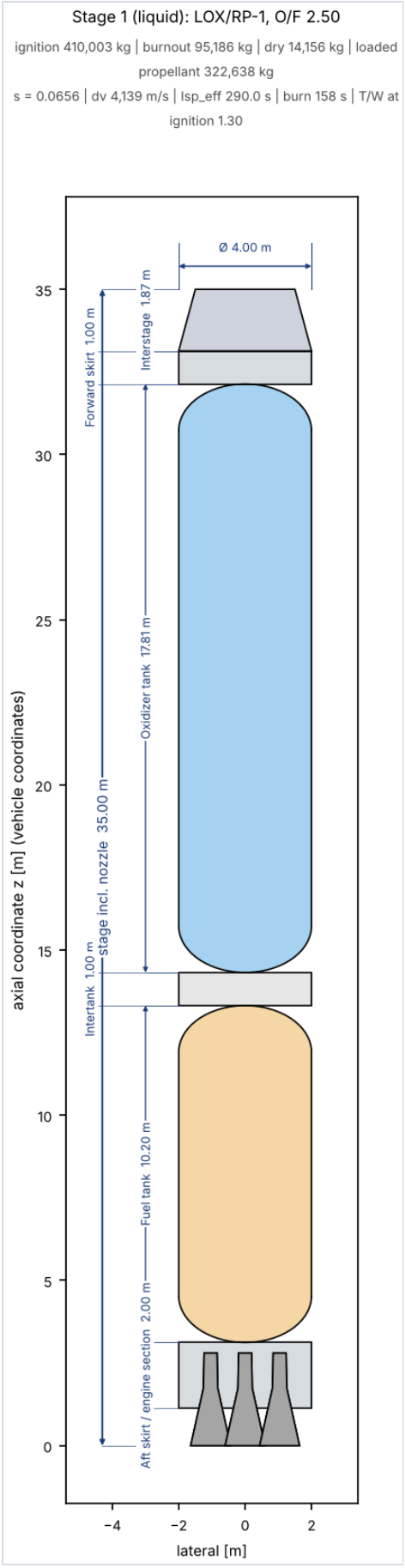


Figure 3. Stage 1 — tankage and engines, to scale. Tank and motor numbers are tabulated in section 7.

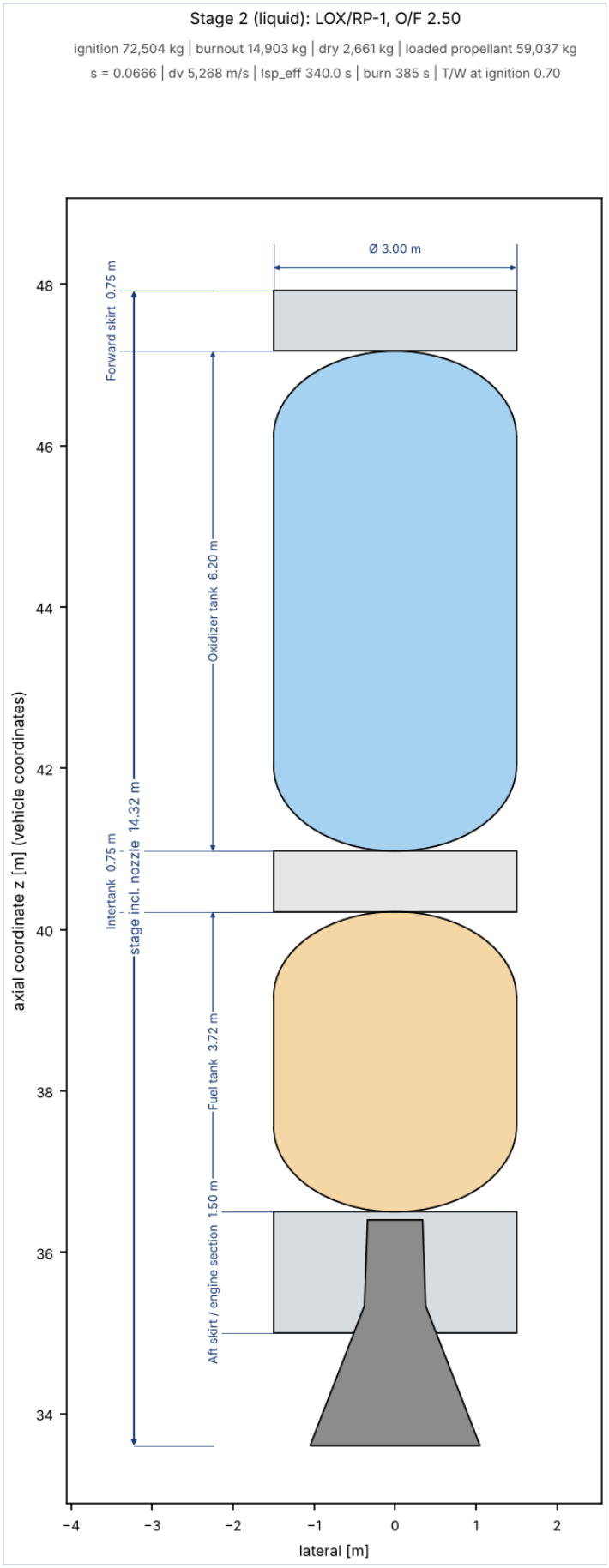


Figure 4. Stage 2 — tankage and engines, to scale. Tank and motor numbers are tabulated in section 7.

4. Propulsion and performance

Stage	T_vac [kN]	T_SL [kN]	Reference condition	mdot [kg/s]	t_burn [s]	T/W ignition	T/W burnout	a_max [g]	mass ratio μ	Δv recomputed [m/s]
1	5,807.8	5,227.0	sea_level	1,974.1	158.5	1.30	6.22	6.22	4.2867	4,139.4
2	497.7	-	vacuum	149.3	384.9	0.70	3.41	3.41	4.8552	5,268.3

The last column recomputes Δv from the sized burn-start and burnout masses with the rocket equation; it must match the allocated Δv in section 2 to solver tolerance.

5. Payload capability

The sections above size a vehicle **for one mission**. This one freezes that vehicle - every stage keeps its dry mass, propellant load, mass flow, burn time and Isp - and asks what payload it delivers to other orbits. Only the payload and its adapter change, so the curves are the vehicle's performance, not a family of re-optimised designs. The design point is marked and lies on the curve by construction, which is also the check that this section and the sizing agree.

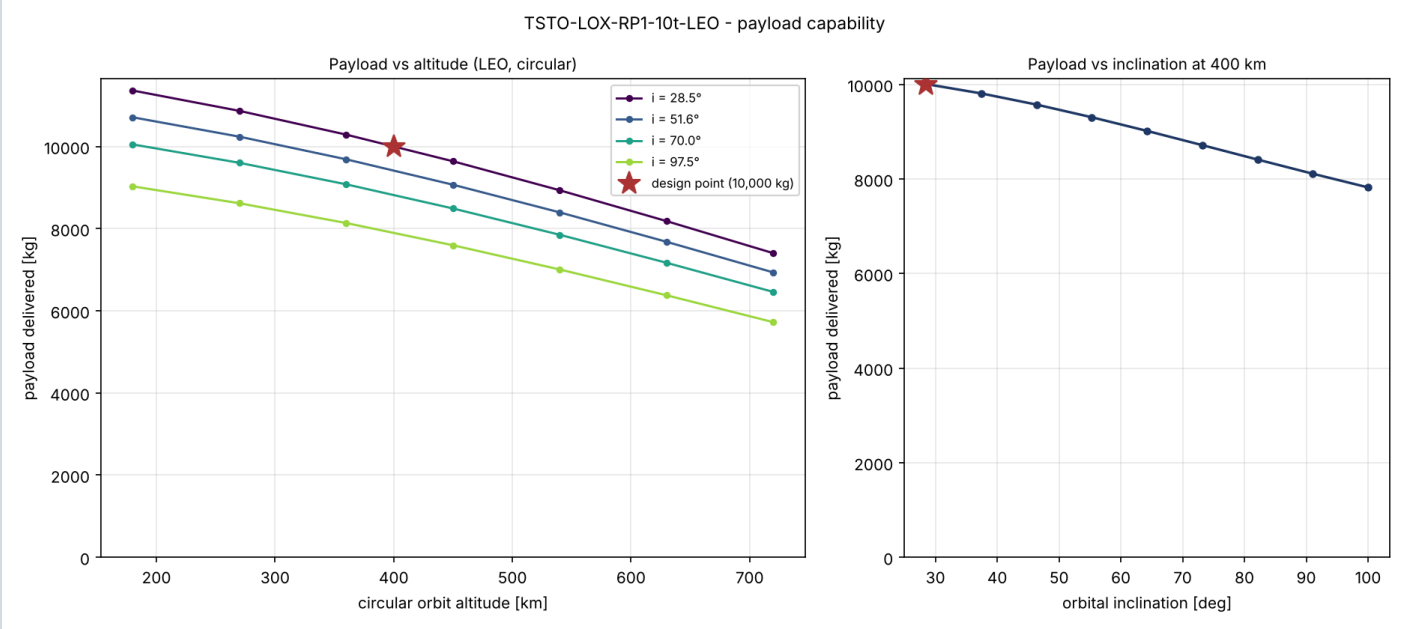


Figure P. Payload against circular-orbit altitude for several inclinations, and against inclination at the design altitude (book Sec. 3.8, Figs. 3.27-3.29). The star is the design point.

Payload [kg] to a circular orbit

Inclination	180 km	270 km	360 km	450 km	540 km	630 km	720 km
28.5°	11,362	10,866	10,288	9,639	8,933	8,182	7,401
51.6°	10,710	10,238	9,687	9,068	8,395	7,679	6,933
70.0°	10,050	9,602	9,078	8,490	7,849	7,168	6,458
97.5°	9,029	8,617	8,136	7,594	7,005	6,377	5,722

Payload against inclination at 400 km

Inclination	Payload [kg]	Note
28.5°	10,007	
37.4°	9,810	
46.4°	9,573	

Inclination	Payload [kg]	Note
55.3°	9,305	
64.2°	9,017	
73.2°	8,716	
82.1°	8,411	
91.1°	8,111	
100.0°	7,822	

An inclination below the launch-site latitude is reported as unreachable rather than flown: the dog-leg or plane change that would reach it is not modelled. The sweep stops at 1,000 km because the book's ascent-loss regressions are fitted to low Earth orbit and extrapolate badly above it.

6. Mass statement

Stage	Carried above [kg]	Mission prop. [kg]	Start-up [kg]	Residual [kg]	Loading margin [kg]	Loaded prop. [kg]	Dry [kg]	Inert at burnout [kg]	Ignition [kg]	Burnout [kg]
1	73,209	312,843	1,974.1	6,256.9	1,564.2	322,638	14,156	21,977	410,003	95,186
2	10,805	57,452	149.3	1,149.0	287.3	59,037	2,661	4,098	72,504	14,903
Stack		**370,295**	**2,123.4**	**7,405.9**	**1,851.5**	**381,675**	**16,817**		**410,003**	

"Carried above" is everything above the stage at its ignition — payload, adapter, the fairing while it is still attached, and every upper stage fully loaded. "Inert at burnout" is dry + residual + loading margin, the mass the structural factor s uses.

GLOM build-up

Item	Mass [kg]
Payload	10,000
Payload adapter	805.0
Fairing	705.3
Stage 1 dry	14,156
Stage 1 propellant loaded	322,638
Stage 2 dry	2,661
Stage 2 propellant loaded	59,037
GLOM	**410,003**
Closure against the sized GLOM	+0.000 kg

Dry-mass breakdown by component

Stage 1

Component	Mass [kg]	% of dry
Engines	5,224.0	36.9
Tank structure (oxidizer)	2,465.9	17.4
Tank structure (fuel)	1,364.3	9.6
Thrust structure	1,332.9	9.4
Mass growth margin	1,286.9	9.1
Pressurant system	645.3	4.6
Aft skirt	334.3	2.4
Interstage	282.5	2.0
Tank insulation (oxidizer)	257.3	1.8
Engine gimbals	199.8	1.4
Intertank	167.1	1.2

Component	Mass [kg]	% of dry
Forward skirt	167.1	1.2
Feed system	150.0	1.1
Other equipment	100.0	0.7
Avionics	70.0	0.5
Separation hardware	60.0	0.4
Wiring	48.5	0.3
Stage dry mass	**14,155.8**	**100.0**

Stage 2

Component	Mass [kg]	% of dry
Engines	615.4	23.1
Tank structure (oxidizer)	451.2	17.0
Avionics	280.0	10.5
Tank structure (fuel)	249.6	9.4
Mass growth margin	241.9	9.1
Aft skirt	188.0	7.1
Thrust structure	126.9	4.8
Pressurant system	118.1	4.4
Intertank	94.0	3.5
Forward skirt	94.0	3.5
Tank insulation (oxidizer)	68.9	2.6
Other equipment	60.0	2.3
Wiring	28.4	1.1
Feed system	25.0	0.9
Engine gimbals	19.7	0.7
Stage dry mass	**2,661.3**	**100.0**

7. Mass ledger and mass properties

Every mass in the vehicle, where it sits on the axis, which shape it was given for inertia, and the rule it came from. This table is the input to the centre-of-mass and inertia calculation and to Figure 2.

▼ Full mass ledger (44 items)

Stage	Group	Component	Mass [kg]	z_cm [m]	Kind	Inertia shape	Source / assumption
1	Propulsion	Engines	5,224.0	1.400	dry	point masses on engine ring (Table 8.15: point mass)	Table 8.9 MER; mid-length of engine envelope
1	Propulsion	Engine gimbals	199.8	2.120	dry	point masses on engine ring	MER; aft skirt mid
1	Propulsion	Thrust structure	1,332.9	2.120	dry	thin cylinder r=R/2, h=D/2 (assumed)	MER; aft skirt mid
1	Tanks	Tank structure (fuel)	1,364.3	8.220	dry	thin shell of revolution (numerical)	tank method per tank spec; shell CM from outline integration
1	Tanks	Tank structure (oxidizer)	2,465.9	23.228	dry	thin shell of revolution (numerical)	tank method per tank spec; shell CM from outline integration
1	Tanks	Tank insulation (oxidizer)	257.3	23.228	dry	thin shell of revolution (numerical)	tank method per tank spec; shell CM from outline integration
1	Structure	Aft skirt	334.3	2.120	dry	thin cylinder (Table 8.10)	13.3 kg/m^2 x area (Table 8.8); thin cylinder (Table 8.10)
1	Structure	Intertank	167.1	13.820	dry	thin cylinder (Table 8.10)	13.3 kg/m^2 x area (Table 8.8); thin cylinder (Table 8.10)

Stage	Group	Component	Mass [kg]	z_cm [m]	Kind	Inertia shape	Source / assumption
1	Structure	Forward skirt	167.1	32.635	dry	thin cylinder (Table 8.10)	13.3 kg/m^2 x area (Table 8.8); thin cylinder (Table 8.10)
1	Structure	Interstage	282.5	34.024	dry	thin frustum (Table 8.10)	13.3 kg/m^2 x frustum area (Table 8.8); CM per Table 8.10
1	Systems	Avionics	70.0	32.635	dry	thin ring (Table 8.15/8.16)	MER/assumption; forward skirt mid
1	Systems	Wiring	48.5	18.061	dry	thin bar along stage (Table 8.17)	1.43 kg/m; mid-stage (Table 8.15: 1/2 l)
1	Propulsion	Feed system	150.0	2.120	dry	point (on axis)	assumption; aft skirt mid
1	Systems	Pressurant system	645.3	32.635	dry	point (on axis)	assumption; forward skirt mid
1	Structure	Separation hardware	60.0	35.001	dry	thin ring	assumption; top of stage
1	Systems	Other equipment	100.0	18.061	dry	point (on axis)	assumption; mid-stage
1	Margin	Mass growth margin	1,286.9	10.223	dry	scaled copy of stage components	single margin; distributed like the stage's other dry components
1	Propellant	Propellant loaded (fuel, RP-1)	92,182.3	8.080	prop_loaded	liquid solid of revolution (numerical)	liquid column centroid, full load
1	Propellant	Residual + margin propellant (fuel)	2,234.6	3.515	prop_resid	liquid solid of revolution (numerical)	liquid at tank bottom at burnout
1	Propellant	Propellant loaded (oxidizer, LOX)	230,455.7	22.867	prop_loaded	liquid solid of revolution (numerical)	liquid column centroid, full load
1	Propellant	Residual + margin propellant (oxidizer)	5,586.5	14.860	prop_resid	liquid solid of revolution (numerical)	liquid at tank bottom at burnout
2	Propulsion	Engines	615.4	35.001	dry	point masses on engine ring (Table 8.15: point mass)	Table 8.9 MER; mid-length of engine envelope
2	Propulsion	Engine gimbals	19.7	35.751	dry	point masses on engine ring	MER; aft skirt mid
2	Propulsion	Thrust structure	126.9	35.751	dry	thin cylinder r=R/2, h=D/2 (assumed)	MER; aft skirt mid
2	Tanks	Tank structure (fuel)	249.6	38.362	dry	thin shell of revolution (numerical)	tank method per tank spec; shell CM from outline integration
2	Tanks	Tank structure (oxidizer)	451.2	44.071	dry	thin shell of revolution (numerical)	tank method per tank spec; shell CM from outline integration
2	Tanks	Tank insulation (oxidizer)	68.9	44.071	dry	thin shell of revolution (numerical)	tank method per tank spec; shell CM from outline integration
2	Structure	Aft skirt	188.0	35.751	dry	thin cylinder (Table 8.10)	13.3 kg/m^2 x area (Table 8.8); thin cylinder (Table 8.10)
2	Structure	Intertank	94.0	40.597	dry	thin cylinder (Table 8.10)	13.3 kg/m^2 x area (Table 8.8); thin cylinder (Table 8.10)
2	Structure	Forward skirt	94.0	47.546	dry	thin cylinder (Table 8.10)	13.3 kg/m^2 x area (Table 8.8); thin cylinder (Table 8.10)
2	Systems	Avionics	280.0	47.546	dry	thin ring (Table 8.15/8.16)	MER/assumption; forward skirt mid
2	Systems	Wiring	28.4	41.461	dry	thin bar along stage (Table 8.17)	1.43 kg/m; mid-stage (Table 8.15: 1/2 l)
2	Propulsion	Feed system	25.0	35.751	dry	point (on axis)	assumption; aft skirt mid
2	Systems	Pressurant system	118.1	47.546	dry	point (on axis)	assumption; forward skirt mid
2	Systems	Other equipment	60.0	41.461	dry	point (on axis)	assumption; mid-stage
2	Margin	Mass growth margin	241.9	40.414	dry	scaled copy of stage components	single margin; distributed like the stage's other dry components

Stage	Group	Component	Mass [kg]	z_cm [m]	Kind	Inertia shape	Source / assumption
2	Propellant	Propellant loaded (fuel, RP-1)	16,867.8	38.312	prop_loaded	liquid solid of revolution (numerical)	liquid column centroid, full load
2	Propellant	Residual + margin propellant (fuel)	410.4	36.694	prop_resid	liquid solid of revolution (numerical)	liquid at tank bottom at burnout
2	Propellant	Propellant loaded (oxidizer, LOX)	42,169.5	43.950	prop_loaded	liquid solid of revolution (numerical)	liquid column centroid, full load
2	Propellant	Residual + margin propellant (oxidizer)	1,025.9	41.233	prop_resid	liquid solid of revolution (numerical)	liquid at tank bottom at burnout
0	Payload	Payload	10,000.0	50.171	payload	solid cylinder (Table 8.17), uniform density assumed	envelope mid-length; uniform solid cylinder assumed
0	Payload	Payload adapter (PAF)	805.0	48.296	paf	thin cylinder (assumed)	PAF mid-length
upper	Fairing	Payload fairing (cylinder)	495.1	49.896	fairing	thin cylinder (Table 8.10)	area-weighted share; thin cylinder (Table 8.10)
upper	Fairing	Payload fairing (nose cone)	210.2	52.871	fairing	thin cone (Table 8.10)	area-weighted share; thin cone, CM at h/3 (Table 8.10)
		Total	**419,260**				

Centre of mass and inertia through the flight events

Event	Mass [kg]	z_cm [m]	z_cm/L	J_roll [kg m²]	J_pitch [kg m²]	J_roll/J_pitch	Note
Liftoff (stage 1 ignition, full load)	410,003	22.788	0.415	742,046	62,654,842	0.0118	
Stage 1 burnout (before separation)	95,186	35.921	0.655	118,455	21,517,929	0.0055	
Stage 2 ignition (after stage 1 separation, fairing jettisoned)	72,504	43.415	0.791	77,330	1,175,855	0.0658	
Stage 2 burnout (before separation)	14,903	47.341	0.863	14,848	375,908	0.0395	
Orbit insertion (payload + adapter)	10,805	50.031	0.912	10,924	15,619	0.6994	

Component locations/shapes are preliminary lumped assumptions (see ledger 'shape' column). Inertias use the book's parallel-axis method; thrust structure, feed system, payload and PAF shapes are assumptions; products of inertia are zero by the axisymmetry assumption.

8. Geometry

▼ Segment table (13 segments, bottom to top)

Stage	Kind	Component	z bottom [m]	z top [m]	Length [m]	D bottom [m]	D top [m]
1	aft_skirt	Aft skirt / engine section 1	1.120	3.120	2.000	4.000	4.000
1	tank	Fuel tank 1	3.120	13.320	10.200	4.000	4.000
1	intertank	Intertank 1	13.320	14.320	1.000	4.000	4.000
1	tank	Oxidizer tank 1	14.320	32.135	17.815	4.000	4.000
1	fwd_skirt	Forward skirt 1	32.135	33.135	1.000	4.000	4.000
1	interstage	Interstage 1-2	33.135	35.001	1.866	4.000	3.000
2	aft_skirt	Aft skirt / engine section 2	35.001	36.501	1.500	3.000	3.000
2	tank	Fuel tank 2	36.501	40.222	3.721	3.000	3.000
2	intertank	Intertank 2	40.222	40.972	0.750	3.000	3.000
2	tank	Oxidizer tank 2	40.972	47.171	6.198	3.000	3.000
2	fwd_skirt	Forward skirt 2	47.171	47.921	0.750	3.000	3.000
-	fairing_cyl	Fairing cylinder	47.921	51.871	3.950	3.000	3.000
-	fairing_nose	Fairing nose cone	51.871	54.871	3.000	3.000	0.000

Tank / motor-case geometry (book Eqs. 8:13-8:16, Table 8.2; solids Eqs. 8:23-8:24)

Stage	Role	Propellant	V internal [m³]	V liquid [m³]	D internal [m]	Dome AR	Barrel L [m]	Dome h [m]	External L [m]	External area [m²]	Tank mass [kg]	Insulation [kg]	Mass method	Bulkhead
1	fuel	RP-1	115.790	112.417	3.992	1.414	7.369	1.411	10.200	133.40	1,364.3	0.0	akin_propellant_mass	none
1	oxidizer	LOX	211.098	202.154	3.992	1.414	14.984	1.411	17.815	229.09	2,465.9	257.3	akin_propellant_mass	none
2	fuel	RP-1	21.188	20.571	2.994	1.414	1.598	1.059	3.721	38.01	249.6	0.0	akin_propellant_mass	none
2	oxidizer	LOX	38.627	36.991	2.994	1.414	4.075	1.059	6.198	61.36	451.2	68.9	akin_propellant_mass	none

Overall	Value
Overall length	54.871 m
Maximum diameter	4.000 m
Maximum span (incl. boosters)	4.000 m
Length / diameter	13.72
Payload envelope	z = 48.671 to 51.671 m

9. Uncertainty

Every mass above is a point estimate from relationships fitted to a historical population. Re-sizing the vehicle 300 times with each MER coefficient drawn from its assumed scatter gives the band below. **Read the band, not the kilogram.**

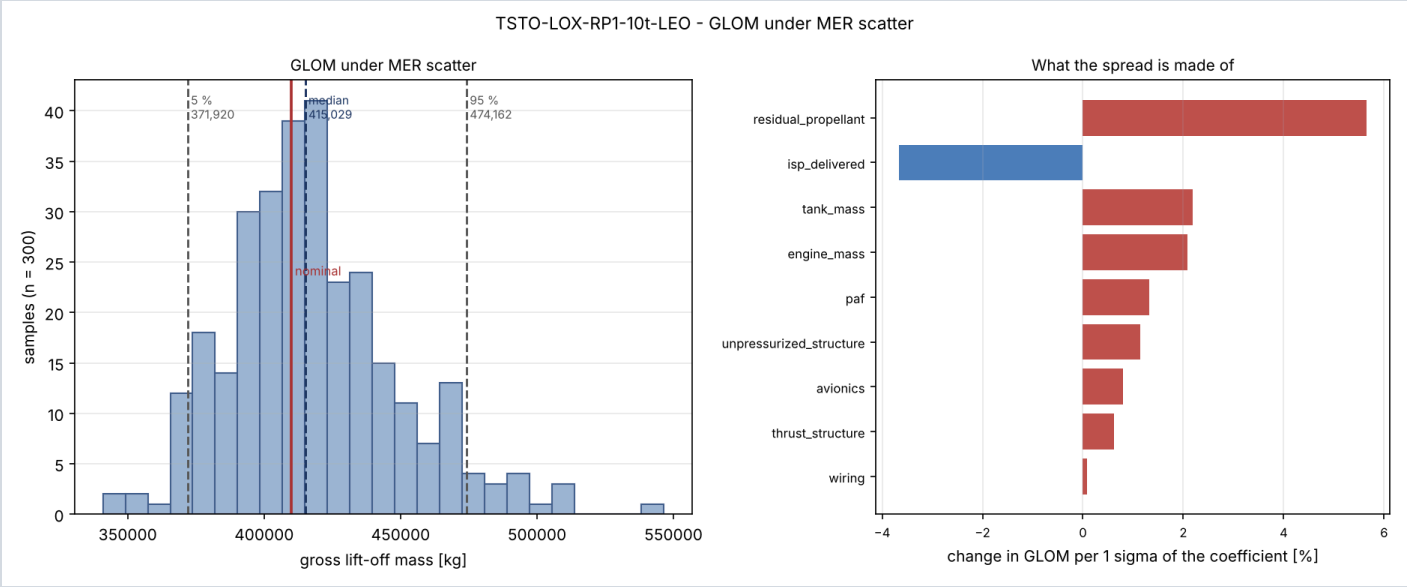


Figure U. Gross lift-off mass under MER scatter, and the one-at-a-time sensitivity that says which coefficient the answer depends on.

Quantity	GLOM [kg]	Relative to nominal
Nominal GLOM (the number in section 5)	410,003	-
Median of the samples	415,029	+1.2 %
5th percentile	371,920	-9.3 %
95th percentile	474,162	+15.6 %
Standard deviation	31,527	7.7 %

Quantity	GLOM [kg]	Relative to nominal
Samples converged	300 of 300	0 failed

Quote this design as 410,003 kg with a 90 % interval of 371,920 to 474,162 kg (-9 % / +16 %), and remember that the interval covers MER scatter only - not a trajectory, loads or a structural concept.

What the spread is made of

MER group	Assumed 1σ	Change in GLOM per +1σ
residual propellant	30 %	+5.66 %
isp delivered	1 %	-3.66 %
tank mass	18 %	+2.19 %
engine mass	15 %	+2.08 %
paf	20 %	+1.33 %
unpressurized structure	20 %	+1.15 %
avionics	30 %	+0.79 %
thrust structure	20 %	+0.62 %
wiring	25 %	+0.08 %

The sigmas are **assumptions**, listed in `config/engineering_assumptions.yaml → mer_scatter` and in the input register: avionics 30 %, engine mass 15 %, isp delivered 1 %, paf 20 %, residual propellant 30 %, solid casing 18 %, tank mass 18 %, thrust structure 20 %, unpressurized structure 20 %, wiring 25 %. Nobody has published a covariance for the book's mass-estimating relationships; these are engineering judgement anchored on this tool's own validation, where the median error against twelve real vehicles is 17 %. Correlations between groups are not modelled, which makes the band, if anything, optimistic.

10. Staging optimum

The book solves the optimum split of the Δv budget between steps in closed form with a Lagrange multiplier (Eqs. 5:30-5:33, pp.287-288), on the assumption that each step's structural ratio and exhaust speed are fixed. In this tool they are not - a stage's structural factor falls out of a mass build-up that depends on its own size - so the two answers differ, and the difference is a measure of how much that assumption costs.

Split	stage 1	stage 2	GLOM [kg]
This design, as allocated	44.0 %	56.0 %	410,003
Analytical optimum (Eqs. 5:32-5:33)	40.3 %	59.7 %	401,533
Numerical optimum (full re-size per evaluation)	35.1 %	64.9 %	397,246

Analytical solution

Step	c_eff [m/s]	σ (converged)	μ optimum	Δv [m/s]
1	2,843.9	0.0656	3.7884	3,787.9
2	3,334.3	0.0666	5.3948	5,619.7

Lagrange multiplier λ = 0.000468 s/m (the book prints L = −λ = -0.000468). With equal exhaust speeds and equal structural ratios this reduces to the book's restricted-staging result, Δv/N per step (p.289).

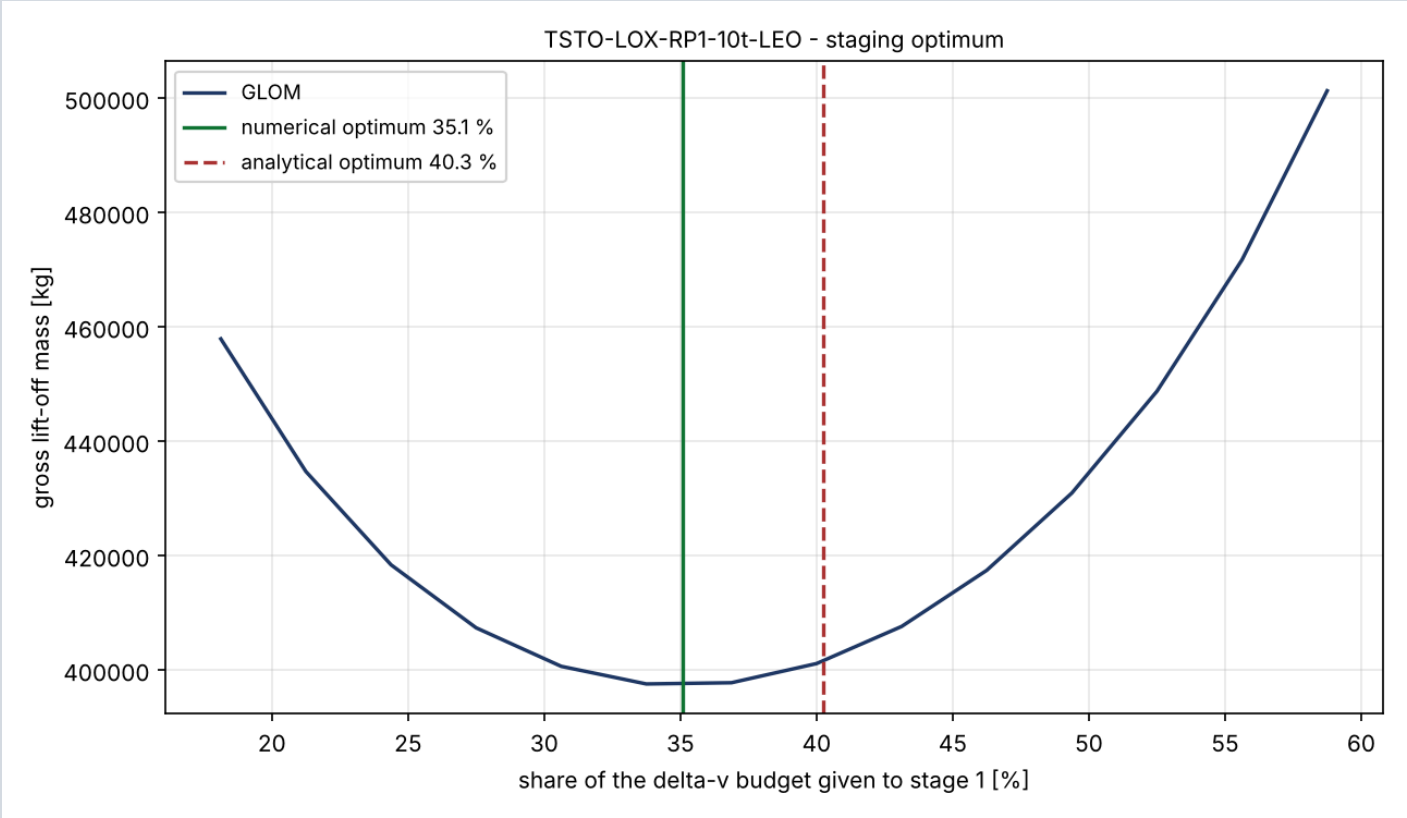


Figure S. Gross lift-off mass against the share of the budget given to stage 1, with both optima marked. A flat curve means the split hardly matters; a sharp one means it does.

11. Input register

Every input the model read, with its units, what it means, where the value came from, and how it is classified. Nothing else was used; nothing here was tuned to make a result match a real vehicle.

user — a requirement or a choice the designer made reference — taken from the book or a cited source

assumed — a placeholder that should be replaced by real data

96 assumed, 105 reference, 38 user, 239 inputs in total.

12. Checks and warnings

Level	Code	Message
WARNING	ACCEL	stage 1 burnout acceleration 6.2 g exceeds limit 6.0 g (throttling not modelled)
INFO	FIDELITY	Preliminary model: losses are user-estimated; no trajectory simulation, loads, stability or thermal analysis. Achievability of the orbit is NOT demonstrated by this tool.

0 errors, 1 warnings. Warnings are not failures: they mark places where the model is being used near or outside the range its methods were fitted for.

13. Engineering method register

Every calculation in the tool, its formulation, the verified book page it comes from (printed page numbers; "SUPPLEMENTARY" means it is **not** claimed to be in the book), its assumptions and limits, where it is implemented, and how it is verified.